

USCD Machine Learning Bootcamp Capstone Results Writeup

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Using several Large Language Models (LLMs), I created an Angular app that tutors its users on the Swedish language for my capstone project. Through sending and receiving prompts from one of four LLMs, my app serves as a user interface that questions the user in Swedish, expects a response, and then provides feedback in English on that response's spelling and grammatical mistakes. The four LLMs supported by the app are Mistral Large 3 [1], Ministral 3B [2], Gemma 3 27B IT [3], and Gemma 3 4B IT [4].

I measure the effectiveness of each using a test dataset of my own creation consisting of 100 Question / Answer / Feedback triplets. The feedback triplets have mostly incorrect Swedish answers with feedback statements consistent with those of an A1 / A2 / B1 learner (eg. incorrect verb tense, missing particles such as "på" or "att", etc.).

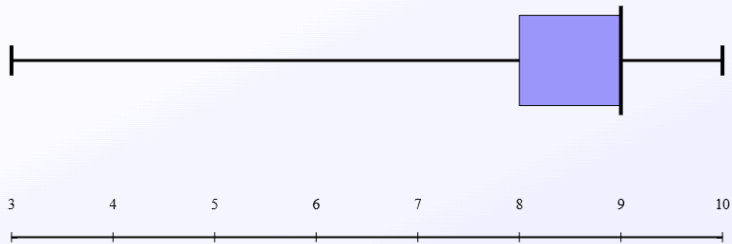
Using the app, I have each LLM provide feedback to the Question and Answer given in each triplet. Then, it compares its feedback to that given by the test set using the following prompt: "Given the AI-provided feedback of [LLM-feedback] and the [feedback-answer-key], please provide a score of 1 to 10 to measure the semantic accuracy of the given AI-provided feedback based upon the given feedback answer key. Please begin the response immediately with the score before going into any reasons or justification." The scores are provided in strings, and the last sentence makes the score values easier to extract and parse into floating-point numbers. In this way, I obtain 100 scores for each LLM from which my app calculates the mean, median, and standard deviation.

Not only do I have each LLM grade itself, I also have that LLM's feedback scored by the other three. Each column represents a distinct grader. I also provide box-and-whisker charts for each LLM using [5]. In this way, here are the mean, median, and standard deviation of the scores of Mistral Large 3 as graded by all four LLMs:

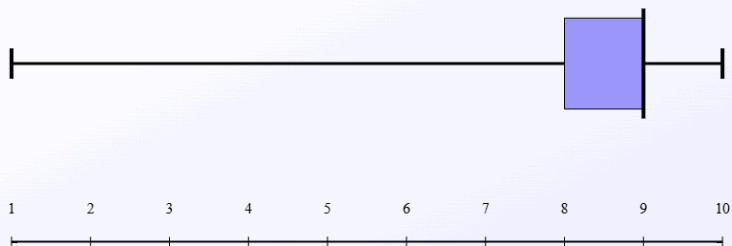
LLM Grader	Mistral Large 3	Ministral 3B	Gemma 3, 27B PT	Gemma 3, 4B IT
Mean	8.19	8.28	7.63	7.61
Median	9	9	9	8
Standard Deviation	1.50	1.80	2.20	1.11

And here are the accompanying box-and-whisker plots on Mistral Large 3's score distributions from each LLM.

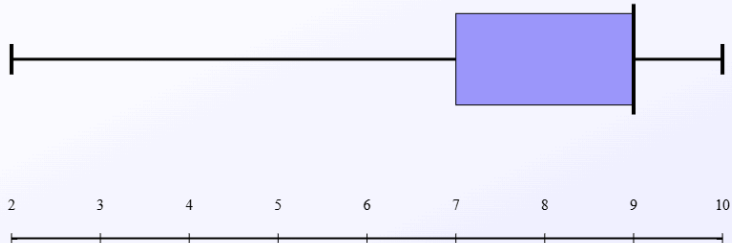
Mistral Large 3
according to Mistral Large 3



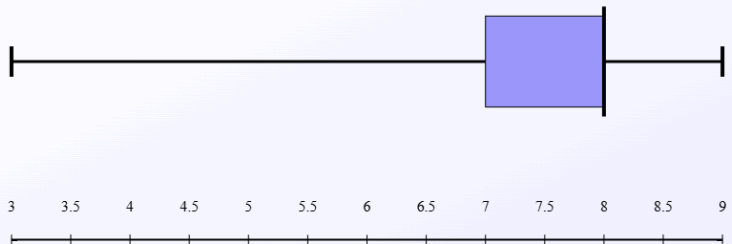
Mistral Large 3
according to Minstral 3B



Mistral Large 3
according to Gemma 3, 27B PT



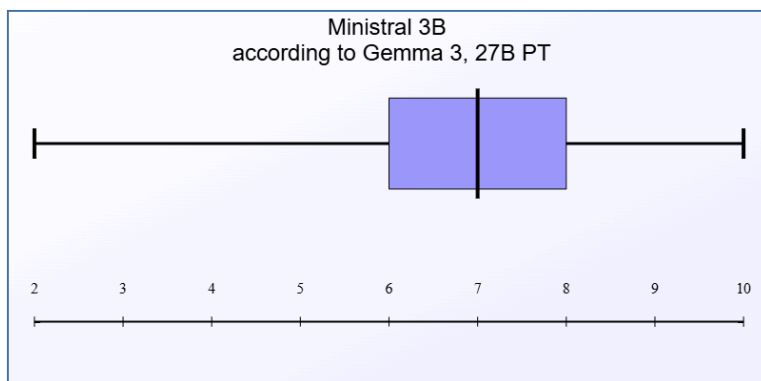
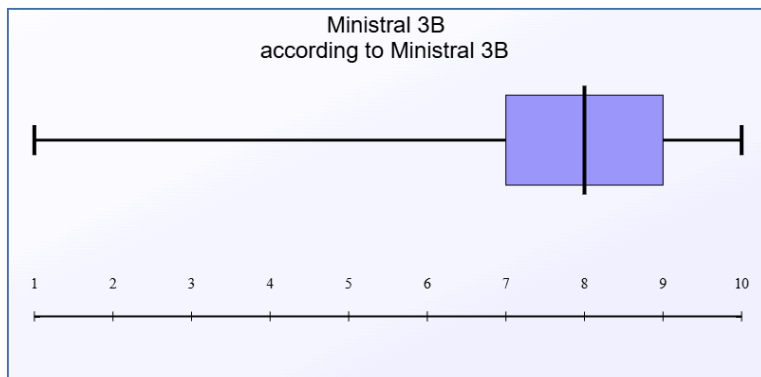
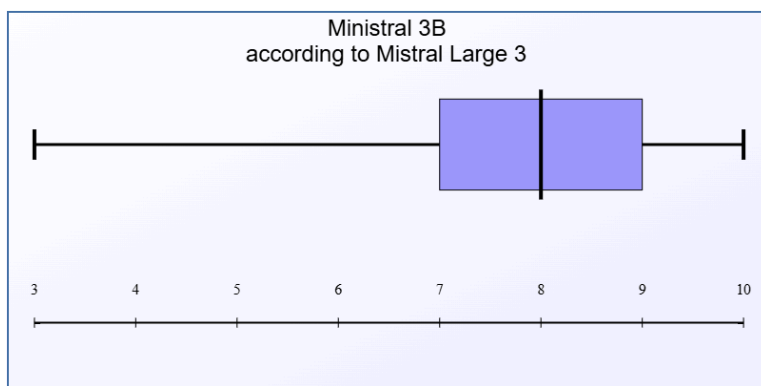
Mistral Large 3
according to Gemma 3, 4B IT

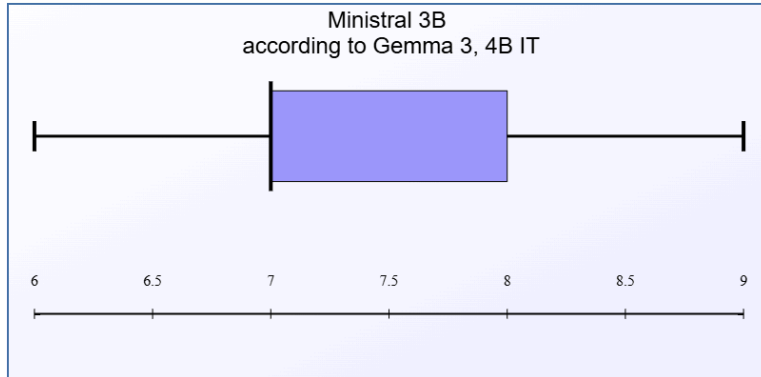


Here are the mean, median, and standard deviation of the scores on Ministral 3B:

LLM Scores	Mistral Large 3	Ministral 3B	Gemma 3, 27B PT	Gemma 3, 4B IT
Mean	7.76	7.91	6.95	7.40
Median	8	8.25	7	7
Standard Deviation	1.40	1.32	1.89	0.82

And here are the accompanying box-and-whisker plots for the score distributions of Ministral 3B:

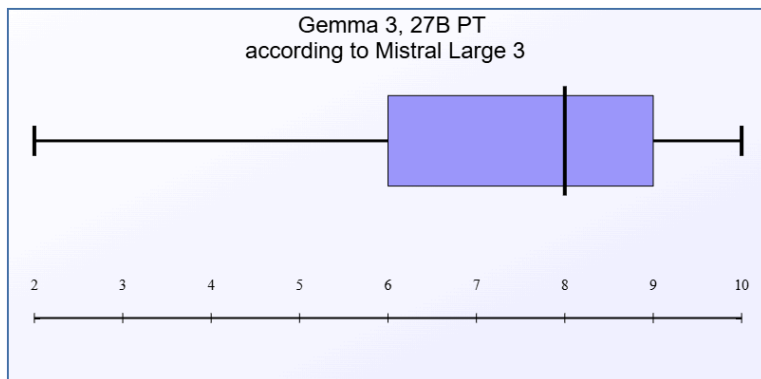


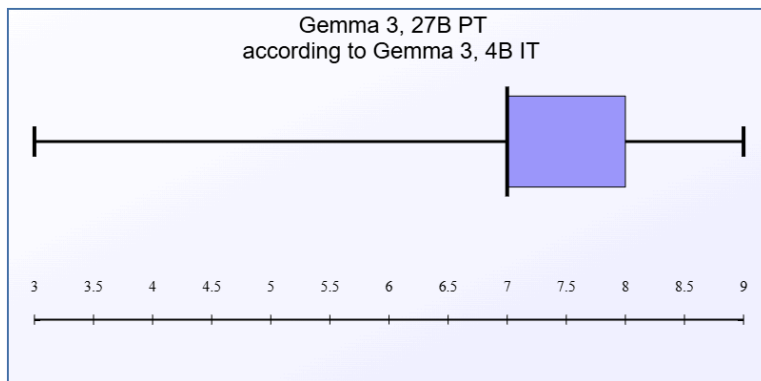
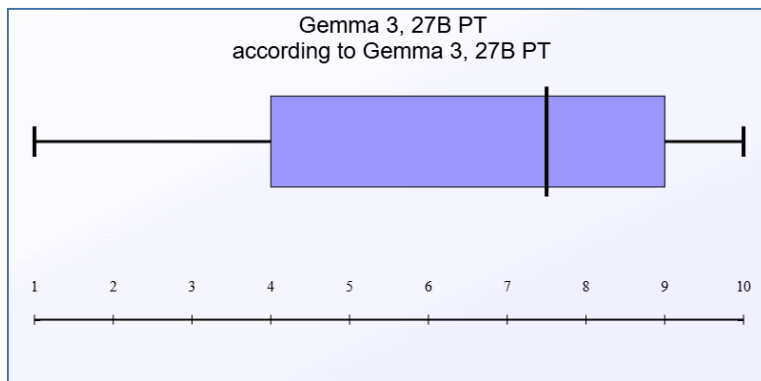
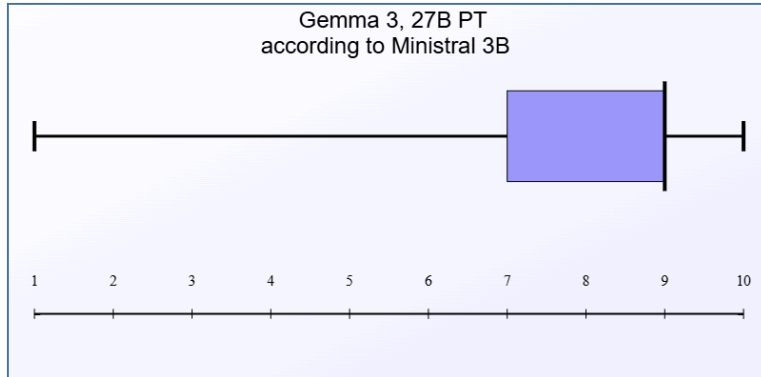


Mean, median, and standard deviation of the scores of Gemma 3, 27B PT are provided here:

LLM Scores	Mistral Large 3	Ministral 3B	Gemma 3, 27B PT	Gemma 3, 4B IT
Mean	7.51	7.90	6.79	7.17
Median	8	9	8	7
Standard Deviation	2.06	2.15	2.51	1.20

And here are the accompanying box-and-whisker plots for the score distributions of Gemma 3, 27B PT:

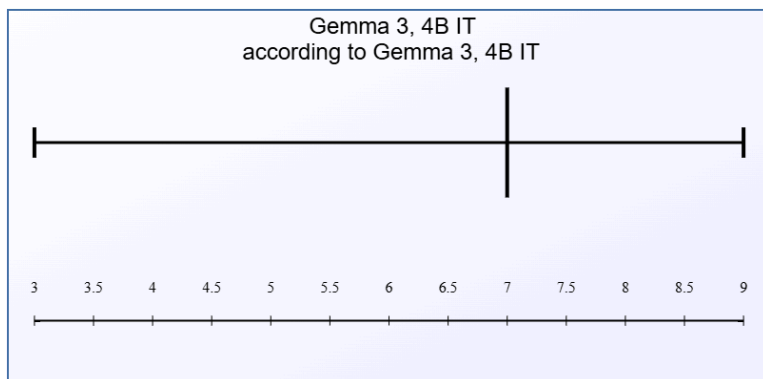
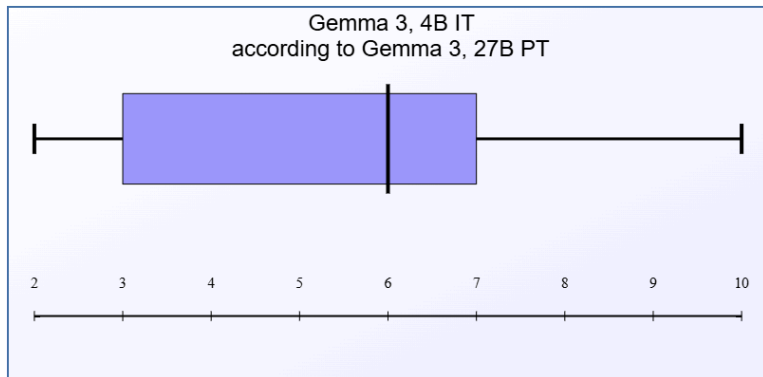
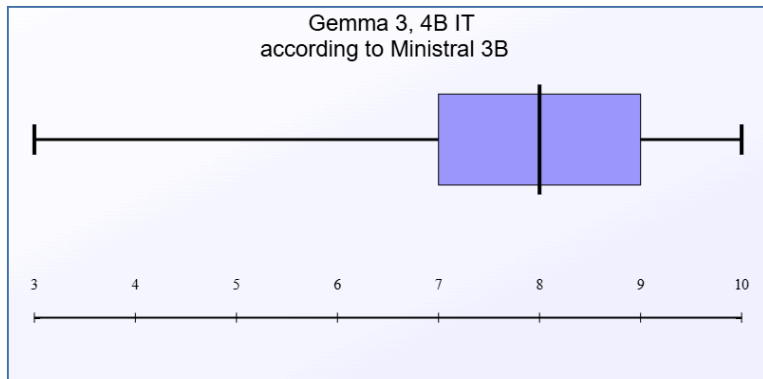
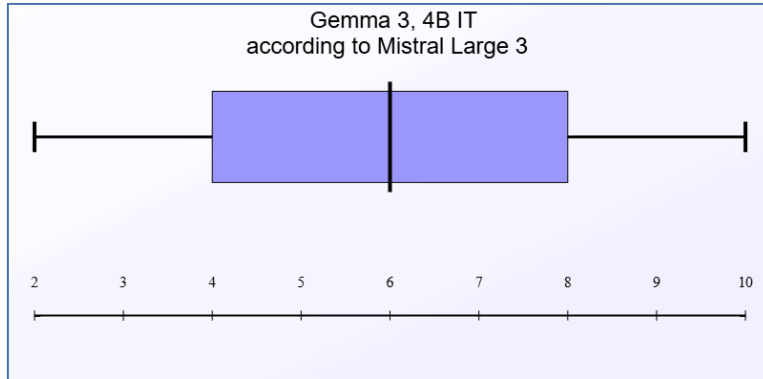




Mean, median, and standard deviation of the scores of Gemma 3, 4B IT are provided here:

LLM Scores	Mistral Large 3	Ministral 3B	Gemma 3, 27B PT	Gemma 3, 4B IT
Mean	6.18	7.75	5.40	6.73
Median	6	8.25	6	7
Standard Deviation	2.00	1.78	2.53	1.12

And here are the accompanying box-and-whisker plots for the score distributions of Gemma 3, 4B IT:



Mistral Large 3 received the highest overall mean and median scores from all four LLMs, thereby making it the highest scoring grader. Minstral 3B gave the highest mean and median scores given to all four LLMs, thereby making it the most forgiving grader.

Mistral Large 3 has 675 billion parameters, so its success is of little surprise unlike that of Ministral 3B. Paradoxically, Ministral 3B scored on par with Gemma 3 27B PT despite only having one-ninth of the number of parameters (3 billion versus 27 billion) with mean scores for both ranging between 6.8 and 7.9. However, Ministral outperforms Gemma 3 27B PT due to the smaller standard deviation of its scores; this suggests a greater consistency in its Swedish comprehension due to a reduced likelihood to completely flub a concept and receive a low score for a number of questions. Moreover, achieving comparable results with fewer parameters indicates that the Mistral series of LLMs are better designed as Swedish tutors compared to the Gemma series.

Ministral 3B's mean scores (ranging between 6.8 and 7.9 across the four LLMs) are less than one point lower than those of Mistral Large 3 (ranging between 7.5 and 8.2) despite having less than one percent of the total number of parameters. How significant is the difference between a "6" and a "7" or between a "7" and an "8"? Let us examine a few examples:

Question: Hur känner spionen sig nu? ("How does the spy feel now?")

Response: Jag tror att spionen verkar lite ledsen. ("I believe that the spy seems a little sad.")

Here was Mistral Large 3's feedback, which it rated as a "6" on a 1 to 10 scale:

"Here's the feedback on the user's response (**"Jag tror att spionen verkar lite ledsen."**) in terms of spelling and grammar: 1. **"Jag"** – Correct spelling and grammar. 2. **"tror"** – Correct spelling and grammar. 3. **"att"** – Correct spelling and grammar. 4. **"spionen"** – Correct spelling and grammar (definite form of **spion**). 5. **"verkar"** – Correct spelling and grammar (present tense of **verka**). 6. **"lite"** – **Grammatical note**: While **"lite"** is correct in many contexts, the more natural choice here would be **"liten"** (masculine/neuter singular form of **liten**), since **"ledsen"** is an adjective describing **"spionen"** (a common gender noun). However, **"lite"** is also acceptable in informal speech. - **Better alternative**: **"Jag tror att spionen verkar lite ledsen."** → **"Jag tror att spionen verkar liten ledsen."** 7. **"ledsen"** – Correct spelling and grammar. **### Overall Assessment**: The sentence is **grammatically correct** and **easily understandable**, though the adjective agreement (**lite** vs. **liten**) could be slightly refined for strict formal correctness. In everyday speech, this would not be considered a mistake. **### Suggested Improvement**: **"Jag tror att spionen verkar liten ledsen."** (More precise adjective agreement) or **"Jag tror att spionen verkar lite ledsen."** (Informal but widely accepted)"

Unfortunately, Mistral Large 3 missed an important detail: "tror" is incorrect because Swedes use "tycker" when speculating on someone else's feelings. Normally, Swedes also use "tror" for speculation but on tangible reality (eg. "I believe it will rain tomorrow" would justify use of "tror" for the verb). Since the rest of the sentence is grammatically correct, and Mistral Large 3 correctly evaluates it as such, "6" seems a fair score here. Take this question and answer pair:

Question: Vad ska du göra i helgen? ("What are you doing this weekend?")

Response: Vi kommer att campa i skogen. ("We are going camping in the forest.")

Mistral Large 3's feedback for this statement was similar to that to the last: everything correctly evaluated except that it missed the wrong verb choice. Both "vi ska" and "vi kommer att" translate to "we will", but Swedes use "ska" when describing intentional future plans whereas "kommer att" is a logical consequence or conclusion of ongoing events. Unlike in the feedback above, Mistral Large 3 rates its own feedback as a "7" rather than a "6". Mistral Large 3 made a similar type of mistake but gave itself an extra point, suggesting that Mistral Large 3 does not view a feedback statement of score "7" to be all that much better than a feedback statement of score "6". Let us examine one more example to show what an "8" looks like according to Mistral Large 3:

Question: Vad rörde du just nu? ("What were you touching just now?")

Response: Jag kände kattens päls. ("I felt the cat's fur.")

Here is Mistral Large 3's "8 out of 10" feedback:

"Here's the feedback on the user's response ***Jag kände kattens päls.*** in relation to the question ***Vad rörde du just nu?*** (*"What were you just touching?"): ### **Mistakes and Corrections:** 1. ***kände*** (incorrect) → ***kände på*** (correct) - ***Kände*** alone means ***felt*** in an emotional or sensory sense (e.g., ***Jag kände glädje*** = ***I felt joy***). - To describe ***physical touch***, Swedish requires the preposition ***på*** (*"kände på" = ***touched/felt***). 2. ***kattens päls*** (grammatically correct but slightly unnatural in this context) - While ***kattens päls*** (*"the cat's fur") is grammatically fine, a more natural phrasing might be ***katten*** (*"the cat") or ***katten på pälsen*** (*"the cat on its fur") if emphasizing the fur. - However, the possessive form (*"kattens päls") is technically correct if the focus is on the fur itself. ### **Corrected Sentence:** - ***Jag kände på katten.*** (*"I touched the cat.") or - ***Jag kände på kattens päls.*** (*"I felt/touched the cat's fur.") ### **Additional Notes:** - The question ***Vad rörde du just nu?*** implies a ***physical action*** (touching something), so ***kände på*** is the appropriate verb phrase. - If the user meant ***I felt the cat's fur*** (as in sensing its texture), the correction still applies. - If the user meant ***I stroked the cat's fur***, a more precise verb would be ***smekte*** (*"stroked") or ***klappade*** (*"petted"). ### **Final Feedback:** The response is ***almost correct*** but needs the preposition ***på*** to convey the intended meaning of physical touch. The most natural version would be: ***Jag kände på katten.*** (*"I touched the cat.")"

Mistral Large 3 correctly identified the missing particle "på". There was one mistake, and Mistral Large 3 found it. This feedback is dramatically better despite being only one point higher.

This suggests that a mean score difference of 1 is often insignificant, but a mean score difference of 2 or more is highly significant and suggests a significantly stronger grasp by the LLM of that linguistic concept. Therefore, Mistral Large 3 and Ministal 3B are the strongest Swedish tutors with Ministal 3B being the most relatively effective for its small parameter size.

The examples also suggest that cultural differences between verbs of similar meanings (eg. “tycker”, “tänker”, and “tror”) are often lost in training even by the most sophisticated models. It is also worth noting that Ministral 3B’s standard deviation is never rated more than 0.5 points higher than that of Mistral Large 3 by the same LLM grader. LLMs can make great learning assistants for users seeking to learn a new language (especially when seeking to learn many new words and build vocabulary), but they have much to learn about the conceptual subtleties languages inherit from culture.

Works Referenced

[1] Mistral Large 3.

<https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-mistral-ai-mistral-large-3.html>

[2] Ministral 3B.

<https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-mistral-ai-ministral-3b.html>

[3] Gemma 3 27B PT.

<https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-google-gemma-3-27b-pt.html>

[4] Gemma 3 4B IT.

<https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-google-gemma-3-4b-it.html>

[5] <https://www.meta-chart.com/box-and-whisker#/your-charts>